Program : **Diploma in Cyber Forensics And Information security**	
Course Code : **5289A**	Course Title: **Data and Application Security Lab**
Semester : **5**	Credits: 1**.5**
Course Category: **Program Elective**	
Periods per week:**3(L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- Presenting symmetric and asymmetric cryptographic systems and covering most important parts of cryptology through introducing many cryptography techniques and algorithms.
- Presenting symmetric and asymmetric cryptographic systems and covering most important parts of cryptology through introducing many cryptography techniques and algorithms.
- Explaining the hash function as an application of cryptography aligning with the concept of message integrity and digital signature authentication.
- Understand the issues involved in using asymmetric encryption to distribute symmetric keys.

Course Prerequisites:

Topic	**Course Code**	**Course Name**	**Semester**
Basic knowledge in Cryptography		Fundamentals of Cryptography and Information Security	III

Course Outcomes:

On completion of the course student will be able to:

COn	**Description**	**Duration (Hours)**	**Cognitive Level**
CO1	Presenting the most important key security requirements that required for any security systems generally and specifically.	9	Applying
CO2	Utilizing and code developing for encryption algorithms that required to achieve confidentiality key security	9	Applying

CO3	Building an appropriate encrypting system that designed for specific key size and message length	9	Applying
CO4	Investigating the suitability of a hash function for verifying the message integrity and digital signature authentication.	15	Applying
	Lab Test	3	

CO – PO Mapping

Course Outcomes	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**
CO1	3		3				
CO2	3		3				
CO3	3		3				
CO4	3		3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	**Name of Experiment**	**Duration (Hours)**	**Cognitive Level**
CO1	**Presenting the most important key security requirements that required for any security systems generally and specifically.**		
M1.01	Introduction to Matlab (Plaintext to Ciphertext as an example. Plaintext encrypting using *strfind()*	3	Applying
M1.02	Introducing the four input arrays (plaintext, alphabet, key, and expected ciphertext) Caesar Algorithm + Mod function	3	Applying
M1.03	Monoalphabetic Ciphering (Shifting position) Playfair Ciphering (Dealing with 5 by 5 metrics + metrics indexes)	3	Applying
CO2	**Utilizing and code developing for encryption algorithms that required to achieve confidentiality key security**		
M2.01	Hill Ciphering (Dealing with metrics operation), Polyalphabetic Ciphering	3	Applying
M2.02	Vigenere Ciphering, Vernam Ciphering	3	Applying

M2.03	One Time Pad Ciphering, Rail Fence Ciphering (Transposition Ciphering)	3	Applying
	Lab Test – I	1.5	
CO3	**Building an appropriate encrypting system that designed for specific key size and message length**		
M3.01	Data Encryption Standards: Building Functions of Bit Decoding (4 to 16) Functions for Substitution and Permutation	3	Applying
M3.02	Advance Encryption Standards: • Building Functions (Substitute Bits and Transformation) • Building Functions (Shift Rows and Mix Columns)	3	Applying
M3.03	Advance Encryption Standards: • Building Functions (Add Round Key Transformation) • Assembling Functions	3	Applying
C04	**Investigating the suitability of a hash function for verifying the message integrity and digital signature authentication.**		
M4.01	Building function for finding *e* relatively prime to	2	Applying
M4.02	Building function for finding *d value*	2	Applying
M4.03	Assembling functions, building pair keys, test the keys	2	Applying
M4.04	Building Elgamal Encryption system	2	Applying
M4.05	One-way Hash function, Secure Hash Algorithm (SHA 3)	2	Applying
	Open ended projects**	5	
	Lab Test – II	1.5	

****Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

- Decrypting Ceasar Cipher

Hash functions take data as an input and return an integer in the range of possible values into a hash table and then it consistently distributes the data across the entire set of possible hash values. The hash function generates completely different hash values even for similar strings.

- Image Encryption: Encrypt image through various algorithms like Advanced Encryption Standard(AES), DES(Data Encryption Standard), and RSA((Rivest-Shamir-Adleman).

Text / Reference

T/R	Book Title/Author
T1	Cryptography and Network Security: Principles and Practice, Global Edition, 7/E, William Stallings, Pearson
T2	Padmanabhan T R, Shyamala C and Harini N, "Cryptography and Security", Wiley Publications 2011.

Online Resources

Sl.No	Website Link
1	https://www.tutorialspoint.com